

Data Engineer with 5+ years of experience across Healthcare, Manufacturing, and Consulting, specializing in designing batch data processing systems, optimizing Spark jobs, and integrating AI into the SDLC. Recent highlights include saving \$200K/year in compute cost through Spark optimizations. Proficient in Azure, Databricks, AWS, SQL, Python, PySpark, and Apache Spark.

EXPERIENCE

Data Engineer

Oct. 2024 – Present

CVS Health | Dallas, TX

- Working in a **cross-functional team of Software Engineers and Product Managers** to build an application that helps rebate analysts estimate rebate value based on historical claims. Scaling a legacy Excel model to process client contracts worth ~\$23B.
- Built the app backend with 60+ multi-cloud PySpark **ETL/ELT pipelines** in production across **Azure (Data Factory, Functions, Data Lake Storage), GCP (Airflow, BigQuery, GCS)**, and Databricks; 40+ **Python APIs**, a **MySQL DB**, and 30+ **Delta tables**.
- Optimized a critical **Databricks** Spark job from 8+ hours to 30 mins (93% faster) by profiling Spark UI metrics and applying **checkpointing, salting, range-based joins, partitioning, and Z-ordering**. Saved \$200k/year in compute cost.
- Designed and built a data validation framework using **PySpark, SQL, Unity Catalog, REST APIs**, publishing accuracy reports on a Databricks dashboard. Cut the UAT cycle from 15 to 4 days for 4 workstreams, unlocking ~\$8B contract opportunity.
- Migrated 30+ data assets from Hive metastore to Unity Catalog, enforcing **table- and row-level access** through AD groups, de-identifying PII and **masking PHI**. Achieved 100% compliance with enterprise data governance standards.

Data Engineer Intern

May 2023 – Oct. 2024

AGCO Corporation | Champaign, IL

- Built an enterprise data hub on **AWS Redshift**, owning 20+ data assets sourced from 3 legacy systems, forming the foundational data layer supporting enterprise-wide analytics and Gen AI initiatives.
- Orchestrated data pipelines using **AWS Glue workflows, S3, Lambda, SES**, with secure credential management through **AWS SSM Parameter Store, Secrets Manager**, and **IAM** roles.
- Standardized deployment for 10+ engineers by replacing manual AWS provisioning (S3, IAM, Glue, Lambda setup) with CloudFormation, cutting per-pipeline deployment time ~75% (roughly 45 min saved per pipeline).
- Replaced manual Excel reporting with a SharePoint-to-Redshift **ELT** pipeline computing **50+ PL/SQL business metrics** into a C-suite OKR dashboard in **Tableau**, freeing 80+ hours/week (~\$350K+/year) in senior stakeholder time.

Analytics Engineer

Aug. 2020 – Jul. 2022

ZS Associates | Pune, India

- Modeled a **conformed star schema dimensional model** over 15+ source tables at the claim-event grain, with patient, provider, and drug dimensions and 4 fact tables totaling 3.5B+ claims. Enabled patient journey analyses for 4M patients across 5 years.
- Enabled the prediction of cancer progression in 4M patients with 78% accuracy by **engineering 500+ features** from 100M+ claims and clinical biomarker data on Databricks, **working closely with Data Scientists** and SMEs.
- Achieved a 50% reduction (4 hours to 2 hours) in run-time by studying access patterns and optimizing 6 critical queries using distribution keys, sort keys, early filtering, and aggregating using CTEs
- Increased a drug's prescriber base by 800 physicians within 3 months through automated physician-targeting insights for 65K+ oncologists based on their patients' cancer progression, built on AWS with MLflow and Databricks Jobs.

TECHNICAL SKILLS

- Proficient:** Python, PySpark, SQL, Apache Spark, Azure (Data Factory, Functions, ADLS, App Insights, Key Vault), AWS (S3, Lambda, Glue, Redshift, CloudFormation, SSM, EC2, IAM), Databricks (Delta Lake, Unity Catalog), MySQL, Git, Tableau
- Familiar:** GCP (BigQuery, GCS, Cloud Composer / Airflow, Dataproc), Data Platforms (Hadoop/Hive, Apache Kafka, dbt), Databases & BI (PostgreSQL, Cassandra, MongoDB, PowerBI), Orchestration & DevOps (Terraform, Docker, GitHub Actions (CI/CD), Prometheus / Grafana)
- AI Tooling:** Claude (skills, hooks), Model Context Protocol (MCP) servers, LLM-assisted debugging and triage

EDUCATION

M.S., Information Management, University of Illinois at Urbana-Champaign (GPA 4.0/4.0)

Aug. 2022 – May 2024

B.E., Electronics & Telecommunication Engineering, University of Mumbai (GPA 9.5/10)

Aug. 2016 – Oct. 2020